7.8 *Shigella Dysenteriae* (Dysentery Bacillus)

The genus *Shigella*are exclusively parasites of human intestine and other primates. *Shigella dysenteriae* is the causative agent of bacillary dysentery or shigellosis in humans. It is a diarrheal illness which is characterized by frequent passage of bloodstained mucopurulent stools. The four important species of the genus *Shigella* are: *Shigella dysenteriae, Shigella flexneri, Shigella sonnei and Shigella boydii.*

7.8.1 Morphology

Shigella are short, Gram negative rods (0.5µm× 1–3 µm in size). They are non – motile, non – sporing and non – capsulated (Figure 7.12).

7.8.2 Cultural Characteristics

- They are aerobes and facultative anaerobes. Optimum temperature is 37°C and optimum pH – 7.4.
- They can be grown on the following media and show the characteristic colony morphology (Table 7.10 & Figure 7.13).

Table 7.10: Colony morphology of Shigella

Media	Colony Morphology
Nutrient Agar	Colonies are circular, convex smooth and translucent
MacConkey Agar	Colourless colonies
SS – Agar	Colourless colonies

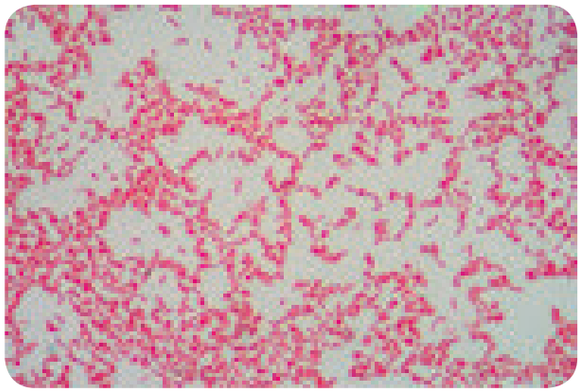


Figure 7.12: Gram staining of *Shigella*



Figure 7.13: Colony morphology of *Shigella* on SS agar

7.8.3 Toxins

Shigella dysenteriae produces toxins, which is of 3 types, namely, endotoxin, exotoxin and verocytotoxin. The mode of action of these toxins is illustrated in the Table 7.11.

7.8.4 Pathogenesis

The pathogenic mechanism of *Shigella dysenteriae*is discussed below in flowchart 7.5.

Source of Infection – Patient or carriers

Route of entry – faecal – oral route

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